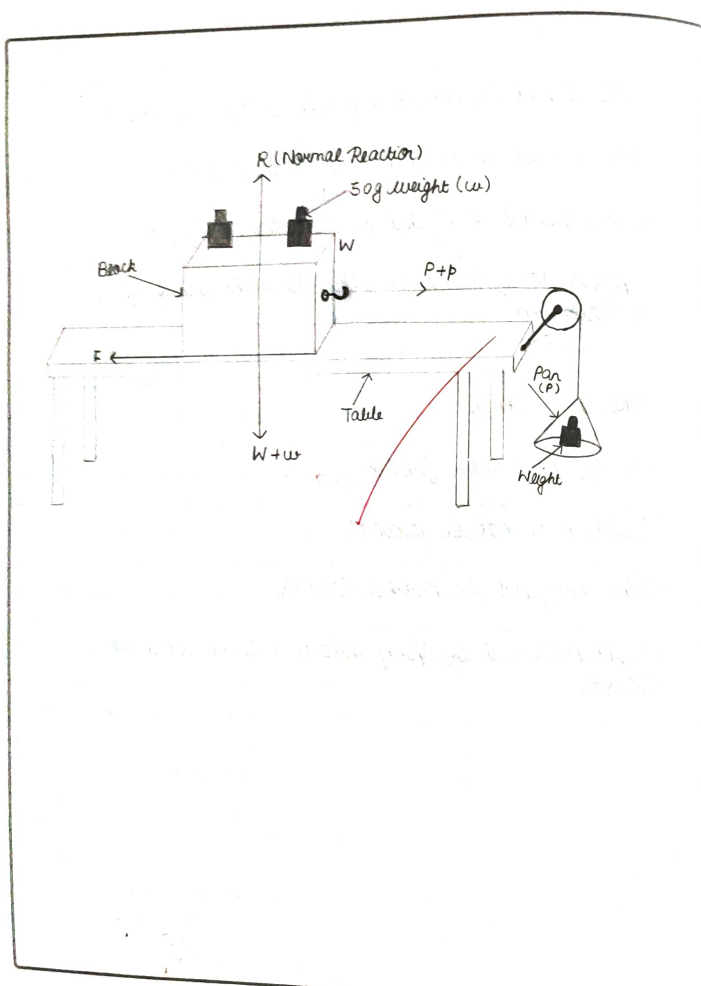




Expt. No. 6

Page No. 28

EXPERIMENT 6

Aim : To study the relationship between force of limiting friction and normal reaction and to find the coefficient of friction between a block and a horizontal surface.

APPARATUS

Wooden Block (with a hook on one side), 50g or 20g weights, horizontal plane (table top) fitted with a frictionless pulley at one end, pan, spring balance, thread, spirit level.

THEORY

Sliding Friction - It is the friction between two surfaces of the bodies in sliding motion.

Force of sliding friction - It is the least force required to make a body start sliding over a surface.

Force of friction, $F \propto R$
 $F = \mu R$

where μ = co-efficient of friction, R is the normal reaction.



Teacher's Signature :

At equilibrium,
and

$$F = P + p$$

$$R = W + w$$

PROCEDURE

1. Clean the horizontal table top and check the pulley to see that it is frictionless (oil if necessary).
2. Weigh the wooden block and put it on the table top. (Check horizontality of table top by spirit level.)
3. Tie one end of a thread with the hook of the wooden block and pass it over the pulley. (The thread must be horizontal).
4. Find the weight of the pan.
5. Tie other free end of the thread with the pan and let the pan hang vertical. (The pan will pull the wooden block horizontally by a force equal to its weight.)
6. Since the pan itself does not pull the block, put some weights in the pan (from weight box).
7. Tap the table top to make the block just slide.



OBSERVATIONS

S.NO	WEIGHT ON BLOCK (W) (gwt)	$R = W + W$ (gwt)	WEIGHT ON PAN (P)	$f = P + P$	$\mu = \frac{f}{R}$
1	0	200	70	90	$90/200 = 0.45$
2	50	250	100	120	$120/250 = 0.48$
3	100	300	130	150	$150/300 = 0.50$
4	150	350	160	180	$180/350 = 0.51$
5	200	400	190	210	$210/400 = 0.52$
6	250	450	160	180	$180/450 = 0.40$
7	300	500	190	210	$210/500 = 0.42$

Here, weight of wooden block (W) = 200gwt
weight of pan = P = 20gwt

Least count of spring balance = 0.2gwt
(2) $\frac{100}{5} = 20\text{gwt}$

Expt. No. 6

Date

Page No. 38

8. Increase weights in pan little by little, till the block just starts sliding on tapping the table top.
9. Note the total weights put in the pan then record them in observation table (sum of weight of pan and weights in pan gives the force of sliding friction).
10. Put one 50g or 20g weight over the wooden block and repeat steps 8 and 9.
11. Repeat steps 8, 9, 10 six times. Every time increase by 50g or 20g.
12. Record the observations in tabular form as given ahead.

RESULT

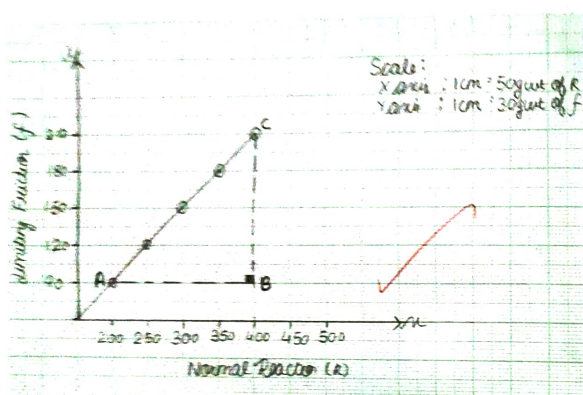
It is found that as the total weight pulled increases, force of limiting friction also increases. The increase is in direct proportion.

The graph shows that limiting friction F is directly proportional to the normal reaction R . It is in agreement with law of limiting friction. (This experiment may be taken as a verification of the law.)



Teacher's Signature : _____

GRAPH



$$\text{slope} = \mu = \frac{BC}{AB} = \frac{210-90}{400-200} = \frac{120}{200}$$

$$\mu = 0.6$$

Expt. No. 6

Date

Page No. 31

The constant ratio F/R , is called coefficient of friction (μ). It can be calculated by finding slope of the graph.

In $\triangle ABC$,

$$\text{slope of straight line } AB = \tan \theta = \frac{BC}{AC} = \frac{F}{R}$$

$$= \frac{210-90}{400-200}$$

$$F = \mu R$$

$$\mu = \tan \theta = 0.6$$

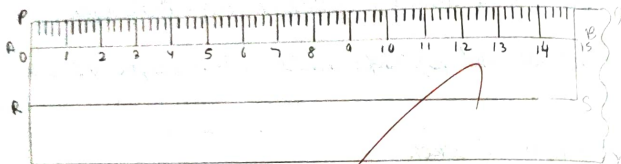
PRECAUTIONS

1. The surface (table top) should be horizontal.
2. The part of thread between block and pulley should be horizontal.
3. Height in pan should be increased in small steps and pan should not oscillate or rotate.
4. Table top should be tapped gently each time.
5. Pulley should be frictionless.

SOURCES OF ERROR

1. The table top may not be horizontal.
2. The thread part between block and pulley may not be horizontal.
3. Pulley may not be frictionless.





Scale of least count 0.2 cm

Exp. No. 37

EXPERIMENT 8

AIM: To make a paper scale of given least count, e.g. 0.2, 0.25, 0.5 cm.

APPARATUS

A thick white paper sheet, pencil, scale with sharp edge marked in cm and mm, file, a pair of scissors, a paper cutter, Thick ivory sheet used by engineering students.

THEORY

- **Least count** - The minimum observation that can be measured by the instrument accurately called the least count of the instrument.
- **Range of an instrument** - The maximum observation that can be measured by the instrument is called its range.

PROCEDURE

1. Fold a white paper sheet in the middle along its length (lengthwise).
2. Mark in the upper half along the length a line PQ (15 cm).

long) by a sharp pencil.

3. Take P as zero mark point on PQ at a distance of 1 cm and write as 0, 1, 2, ... up to 15.
4. Mark the vertical lines to line PQ at the position of each mark 0, 1, 2, ... up to 15.
5. Draw another sharp line RS which is parallel to PQ at a distance of 8 mm.
6. Draw another line XY parallel to PQ at a distance of about 25 mm, and complete the rectangle ABXY.
7. Now, divide each 1 cm interval into five equal divisions on PQ by marking points at every interval of 2 mm. Mark these points upto 15 cm.
8. Next draw sharp small lines about 3 mm long perpendicular to PQ on each of the point which is separated by 2 mm.
9. Draw another line AB parallel to PQ at a gap of 3 mm.
10. Darken each line and division by the sharp black pen and write 1, 2, ... 15 at each cm mark.
11. Cut the rectangular scale by a sharp paper cutter and paste it on a thick, ivory sheet and cut the sheet along.



the sheet of the rectangle with the help of sessions.

2. Paper scale of least count 0.2mm and range of 15cm is ready.

RESULT

1. The scale of least count 0.2cm has been made.

PRECAUTIONS

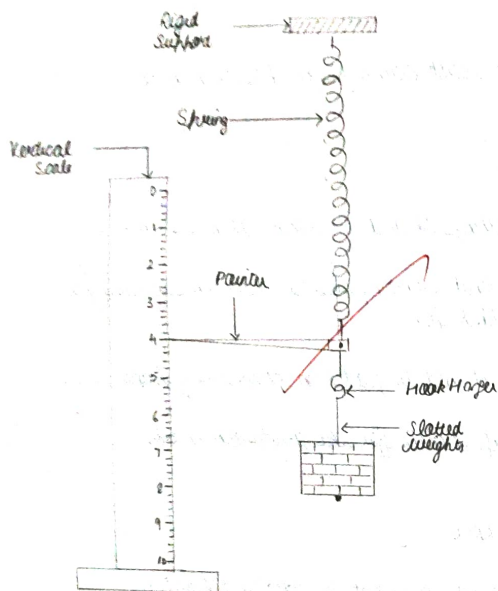
1. The cm markings should be longer than 0.2cm.
2. Final lines and marking should be drawn by using fine tipped black ink pen.
3. Paper scale should be pasted on thick ivory paper.
4. Use very sharp pencil for the graduation marks.

SOURCES OF ERROR

1. Graduation marks may not be equally separated.
2. The lines showing graduation may not be sharp as required.



Teacher's Signature : _____



Extension of spring

Expt. No. 9

Date

Page No. 40

EXPERIMENT 9

AIM: To find the force constant of a helical spring by plotting a graph between load and extension.

APPARATUS

Spring, a rigid support, a 50g or 20g hanger, six 50g or 20g slotted weights, a vertical wooden scale, a fine pointer, a hook.

THEORY

When a load F is suspended from lower free end of a spring hanging from a rigid support, it increases its length by amount l , then

$$F \propto l$$

$$F = Kl$$

where K is constant of proportionality. It is called the force constant or the spring constant of the spring.

From above, if, $l = 1$, $F = K$

Hence, force constant (or spring constant) of a spring may be defined as the force required to produce unit extension in the spring.



Teacher's Signature : _____

OBSERVATIONS

S.No	F (gwt)	$m_1 = m_2 - m_0$ (LOADING)	$m_2 = m_1 - m_0$ (UNLOADING)	$m = \frac{m_1 + m_2}{2}$	$K = \frac{F}{m}$
1	0	0	0	0	0
2	250	$29.5 - 24 = 5.5$	$29.5 - 24 = 5.5$	5.5	$250/5.5 = 45.4$
3	500	$35.5 - 24 = 11.5$	$35.5 - 24 = 11.5$	11.5	$500/11.5 = 43.4$
4	750	$41.5 - 24 = 17.5$	$41.5 - 24 = 17.5$	17.5	$750/17.5 = 42.8$
5	1000	$47.5 - 24 = 23.5$	$47.5 - 24 = 23.5$	23.5	$1000/23.5 = 42.5$

Expt. No. 9

Date

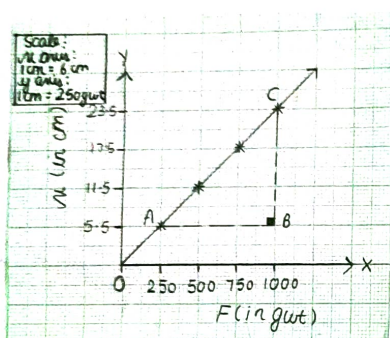
Page No. 31

PROCEDURE

1. Suspend the spring from a rigid support. Attach a pointer and a hook from the lower free end.
2. Hang a 50 g hanger from the hook.
3. Set the vertical wooden scale such that the tip of the pointer comes over the divisions on the scale but does not touch the scale.
4. Note the reading of the position of the tip of the pointer on the scale. Record it in loading column against zero load.
5. Gently add suitable weight of 50 g or 20 g slotted weights to the hanger. The pointer tip moves down.
6. Wait for few minutes till the pointer tip comes to rest. Repeat step 4.
7. Repeat steps 5 and 6 till six slotted weights have been added.
8. Now, remove one slotted weight. The pointer tip moves up. Repeat step 6. Record the reading in unloading column.
9. Repeat step 8 till only hanger is left.
10. Record your observations.

Teacher's Signature

GRAPH



$$K = \frac{F}{x} = \frac{F}{x} = \frac{AB}{BC}$$

$$K = \frac{1000 - 250}{23.5 - 5.5} = 41.67 \text{ gwt/cm}$$

Expt. No. 9

Date

Page No. 42

RESULT

The force constant of the given spring is 41.67 gwt/cm

PRECAUTIONS

1. Loading and unloading of weight must be done gently.
2. Reading should be noted only when tip of the pointer comes to rest.
3. Pointer tip should not touch the scale surface.
4. Loading should not be beyond elastic limit.

SOURCES OF ERROR

1. The support may not be rigid.
2. The slotted weights may not have correct weight.

Teacher's Signature :